

LOW LEVEL DESIGN (LLD)

PREREQUISITE

- ① OOP
- ② Any oop language
- ③ SOLID Design Principle
- ④ Design Patterns
- ⑤ UML Diagram

WHAT IS LLD?

LLD describes how individual classes, objects, methods, and their relationships are designed a system.

[LLD = detailed blueprint of code]

Ex

House

- Bedroom
- Kitchen
- Bathroom

Bedroom

Size = 10 X 10
window count = 2
Door position = left

etc.

Ex E-commerce App

Customer can place order and make payment.

HLD

frontend — backend — database

LLD

```
class Customer {
```

```
    →  
    →  
    →  
}
```

```
class Order {
```

```
    →  
    →  
    →
```

```
}
```

```
interface Payment {
```

```
    →  
    →
```

```
}
```

```
class UPIPayment {
```

```
    →  
    →  
    →
```

```
}
```

- ① Class Diagram
- ② Design pattern & SOLID
- ③ Database Schema
- ④ API design

POINT TO REMEMBER

- * System design does not have a single correct solution. Every system comes with certain trade-offs. If you understand those trade-offs and determine that they can be accepted based on our requirements, we can consider it a good design.
- * Different engineers may design the same system in different ways. If you understand the trade-offs, advantages and disadvantages of your design and can justify your decisions, then it can be considered a good system design.

How To Approach a LLD Question

① Design round

② Machine coding round

Design Round

① 40-45 min approx.

② Problem statement

① Design BMS

② Design Parking lot

Approach

① Overview

② Gather Requirements

→ Clarify Req.

① ask doubts

② Corner cases

④ class diagram

⑤ Schema design

⑥ Write code / answer follow-up questions.

Machine coding Round

- ① 2-2.5 h approx.
- ② Given detailed set of req.

Approach

- ① clarity req.
 - ② class diagram
 - ③ Schema design
 - ④ End to end working code
 - ⑤ Answer follow-up question.
-

Step-1

Overview

- ① Align with interviewer.

Design X

you know the system

- ① Align yourself with the expectation
 - ① tell what you know
 - ② Ask if those are correct.

you don't know the system

- ① Be humble to ask
- ② Interviewer should not feel you are gathering requirements.

- ③ Ask follow up questions

Q1 :- what kind of design interview is asking?

- ① Entity design OR
- ② S/W system design

Q2:- Do I need to persist the data

① In-memory

OR

② Real DB

Q3:- How does someone interact with the system.

- API OR

- CMD OR

- Hardcode

Step-2 Gather & Clarify Req.

- ① In overview phase, build a visualization of the system in your mind. so that you are able to think of req.
- ② Suggest 5-8 core features.
- ③ Note down the req. in a notepad

Step-3

Class diagram

Step-4

Schema Design

Step-5

Write code / answer follow-up questions.